



SISMO SFTP FILE UPLOADER

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SftpClient Setup and Usage Guide - Ubuntu

This guide explains how to prepare, configure, and run SftpClient.jar on Ubuntu. It also includes instructions for scheduling the application to run daily at 05:00 AM and for removing the scheduled task.

Note: 05:00 AM is configured in the local time zone. It is recommended to adjust the schedule so that file uploads occur at 5:00 AM UTC. **Note:** SftpClient is a Java-based application that connects to an SFTP server and uploads files to a specified directory.

Prerequisites

1. **Java Runtime Environment (JRE)** Ensure the jre folder is included in the same directory as the SftpClient.jar.
2. **Environment Variables** The run_ubuntu.sh script requires several environment variables to be set. Update these to match your environment before execution.
3. **Required Folders** The following folders in the root directory of the application (created during the zip packaging) are used by default for file uploads:
 - portfolio: for portfolio files.
 - user_indicator: for stock-related indicator files.
 - macro_indicator: for macro indicator files.

You may create your own folders and update the environment variables in run_ubuntu.sh accordingly.

Environment Variables

Below there are key environment variables, their purposes, and usage examples:

SFTP_SERVER

- **Purpose:** Specifies the hostname or IP address of the SFTP server to which files will be uploaded.
- **Example:** sftp.example.com or 192.168.1.100

SFTP_USER

- **Purpose:** The username used for authenticating with the SFTP server.
- **Example:** sftpuser



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SFTP_PRIVATE_KEY

- **Purpose:** Path to the private SSH key file used for SFTP authentication.
- **Example:** /home/youruser/.ssh/id_rsa

SFTP_PHRASE

- **Purpose:** Passphrase for the private SSH key, if required. Leave empty if the key is not encrypted.
- **Example:** yourSecretPassphrase
(Leave blank if not needed)

SFTP_LOCAL_USER_INDICATOR_DIRECTORY

- **Purpose:** Directory path containing user indicator files to be uploaded.
- **Example:** /home/youruser/sftpclient/user_indicator

SFTP_LOCAL_MACRO_INDICATOR_DIRECTORY

- **Purpose:** Directory path containing macro indicator files to be uploaded.
- **Example:** /home/youruser/sftpclient/macro_indicator

SFTP_LOCAL_PORTFOLIO_DIRECTORY

- **Purpose:** Directory path containing portfolio files to be uploaded.
- **Example:** /home/youruser/sftpclient/portfolio

SFTP_PORTFOLIO_FILE_MAPPER

- **Purpose:** Maps external IDs to substrings in file names for portfolios, which are stored on a user's side, enabling, enabling automatic renaming before upload. There should be no hyphens/dashes in the portfolio names (e.g. Portfolio1 as below)
- **Example:** *PortfolioExternalId1-Portfolio1, PortfolioExternalId2-Portfolio2*
- **Mock File Name:** Portfolio1_Report.csv renamed to PortfolioExternalId1-[timestamp]-[operation].csv (The substring Portfolio1 in the file name matches the mapping PortfolioExternalId1-Portfolio1. The file is renamed to PortfolioExternalId1-[timestamp]-[operation].csv before being uploaded)

SFTP_PORTFOLIO_FILE_MAPPER_DEFAULT_OPERATION

- **Purpose:** Specifies the default operation for portfolio files.
 - F = replace file and delete history
 - M = modify file
- **Example:** M Files will be uploaded with the “modify” (update) operation unless specified otherwise.

SFTP_USER_INDICATOR_FILE_MAPPER

- **Purpose:** Same logic as for portfolios, but applies to user indicator files.
- **Example:** *UserExternalId1-UserIndicator1, UserExternalId2-UserIndicator2*



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- **Mock File Name:** UserIndicator1_Data.csv renamed to UserExternalId1-[timestamp]-[operation].csv and zipped before upload

SFTP_USER_INDICATOR_FILE_MAPPER_DEFAULT_OPERATION

- **Purpose:** Default operation for user indicator files.
 - F = replace file and delete history
 - M = modify file
- **Example:** M Files will be uploaded with the “modify” operation unless specified otherwise.

SFTP_MACRO_INDICATOR_FILE_MAPPER

- **Purpose:** Same mapping logic, applied to macro indicator files.
- **Example:** MacroExternalId1-MacroIndicator1, MacroExternalId2-MacroIndicator2
- **Mock File Name:** MacroIndicator1_Stats.csv renamed to MacroExternalId1-[timestamp]-[operation].csv

SFTP_MACRO_INDICATOR_FILE_MAPPER_DEFAULT_OPERATION

- **Purpose:** Default operation for macro indicator files. F means replace the file, and M means modify the file.
- **Example:** F Files will be uploaded with the “replace” operation unless specified otherwise.

✓ Ensure all environment variables are correctly set in *run_ubuntu.bat* before running the application. If you encounter issues, check the log file (log.txt) in the application’s directory for details.

Manual Execution Steps

1. **Edit the Script**
2. **Make the File Executable:**
 - Open the terminal and navigate to the directory containing *run_ubuntu.sh*.
 - Run the following command to ensure it is executable:
`chmod +x run_ubuntu.sh`
3. **Run the Script:**
 - Execute the script:
`./run_ubuntu.sh`
4. **Verify Execution:**
 - The script will check for the presence of the Java runtime and execute *SftpClient.jar*.



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Schedule SftpClient.jar to Run Daily at 05:00 AM

1. Prepare the Script:

- Ensure schedule_ubuntu.sh is in the same directory as run_ubuntu.sh.

2. Run the Scheduler Script:

- Execute schedule_ubuntu.sh to create a cron job:

```
./schedule_ubuntu.sh
```

3. Verify the Task:

- Check the cron jobs to confirm the task is scheduled:

```
crontab -l
```

Remove the Scheduled Task

1. Run the Removal Script:

- Execute remove_schedule_ubuntu.sh to remove the cron job:

```
./remove_schedule_ubuntu.sh
```

2. Verify Removal:

- Check the cron jobs to confirm the task is removed:

```
crontab -l
```

Schedule SftpClient.jar to Run Daily at 05:00 AM

1. Prepare the Script:

- Ensure schedule_ubuntu.sh is in the same directory as run_ubuntu.sh.

2. Run the Scheduler Script:

- Execute schedule_ubuntu.sh to create a cron task:

```
./schedule_ubuntu.sh
```

3. Verify the Task:

- Check if the task is loaded:

```
crontab -l | grep run_ubuntu.sh
```

Remove the Scheduled Task

1. Run the Removal Script:



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- Execute `remove_schedule_ubuntu.sh` to remove the cron task:

```
./remove_schedule_ubuntu.sh
```

2. Verify Removal:

- Check if the task is unloaded:

```
crontab -l | grep run_ubuntu.sh
```

Important Notes



- All `.sh` scripts (`run_ubuntu.sh`, `schedule_ubuntu.sh`, `remove_schedule_ubuntu.sh`) must be in the same directory as `SftpClient.jar` and the `jre` folder.
- The folders `user_indicator`, `macro_indicator`, and `portfolio` are created during packaging. You may use custom folders by updating `run_ubuntu.sh`.
- If you encounter errors, check logs in the Task Scheduler or in `log.txt`.

File Structure

The directory should have the following structure:

```
/your-directory
├─ run_ubuntu.sh
├─ schedule_ubuntu.sh
├─ remove_schedule_ubuntu.sh
├─ SftpClient.jar
├─ README_UBUNTU.md
├─ README_UBUNTU.pdf
├─ jre/
├─ user_indicator/
├─ macro_indicator/
└─ portfolio/
```

Troubleshooting

- **Java Runtime Not Found** Ensure the `jre` folder is present.
- **Task Scheduler Errors** Confirm you have the required permissions. Check the History tab for details.
- **Incorrect Environment Variables** Double-check the variables in `run_ubuntu.sh` for typos or invalid values.